Short-list for IPT 2027

To stimulate creativity and interesting discussions, all the problems have been formulated as open physics questions with a statement that is not too stringent. Therefore, there is no uniquely favored understanding of the problem conditions and it is up to each team to interpret the conditions in a way that is both interesting from a physical perspective and coherent with the problem statement. It is assumed that every phenomenon will be studied theoretically and, whenever possible, experimentally, to propose and validate a model that explains the phenomenon and its dependence on the most relevant parameters. Teams are encouraged to carefully consider their time budgets, to strike a balance between the number of problems that are studied and the depth with which each problem is investigated.

All experiments should comply with local safety regulations and care should be exercised when dealing with dangerous equipment and substances. Teams are solely responsible for any damage or injuries incurred while working (or thinking) on the problems.

Everyone is allowed to vote! Please use the following form.

To vote, distribute 100 points among the problems you prefer, but try not to spread it between too many problems, as points will be lost in the noise of the other votes. The vote is open until August 30th 2026.

Electromagnetism

1. The magnetic hourglass

Given a confined column of granular material where at least some of the grains are ferromagnetic, an external magnetic field can redirect internal stresses and greatly alter the effective mass of the column. Consider an hourglass containing such a material instead of regular sand. Investigate how an external magnetic field affects the flow rate. What properties of the field (amplitude, gradient, time-variation etc.) can be inferred from observing the granular flow?

2. Alcon levitator II

In the conventional Alcon levitator, a permanent magnet levitates above two counter-rotating, metal cylinders. Explain the phenomenon and investigate how it changes for other conductor shapes, e.g. spheres, or cylinders with polygonal cross-section. With shape as a parameter, maximize the distance between magnet and conductor surface where the levitation is stable. How does the maximum distance of stability changes if the conductor has holes?

3. Electromagnetic Pendulums

Hanging magnets is a rare case of coupled mechanical motion with minimal friction and no collisions, but a metal plate can both dampen and excite the system depending on its relative motion. Investigate the dynamics of one or more magnetic pendulums near a driven, conductive plate, including types and regimes of motion, resonances, and whether chaotic motion is possible. https://www.youtube.com/watch?v=-T0ORDWg-6I

4. Magnetic mergers

When two bodies orbit each other and continuously lose gravitational energy, their separation decreases, their orbital frequency rises, and their inspiral accelerates. This runaway merger process is responsible for the gravitational wave signals detected from inspiralling neutron stars and black holes, where energy escapes as ripples in spacetime.

Two neutrally buoyant magnetic spheres, submerged in a viscous fluid might undergo an analogous inspiral: magnetic attraction binds them, an initial tangential impulse sets them into orbit, and viscous dissipation steadily drains their orbital energy. Investigate this system. One of the magnets may be held fixed. How do the orbital radius and frequency evolve with time? How does the time to merger scale with the initial separation, the fluid viscosity, and the magnetic interaction strength? Can you maximize the number of orbits before merger? To what extent does the viscous magnetic inspiral reproduce the phenomenology of gravitational wave-driven binary inspiral, and where, precisely, does the analogy break down?



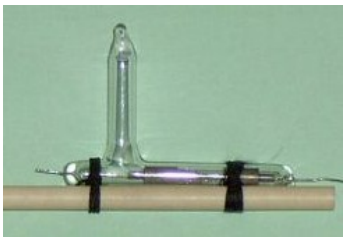
5. Magnetic Chains

A typical chain is made of many small metallic links connected. Such a chain, if long enough, exhibits features that resemble a rope. However, if the links are made of a ferromagnetic material, and you attach one of the ends of the chain to a permanent magnet, the chain will start to behave differently, rather like a malleable material. This means that the chain can be bent and twisted while still maintaining its stability. What is the range of stable shapes that can be formed by the chain, and how does it depend on the length of the chain, the strength of the magnet and the friction between the links? Explain the phenomenon and investigate the relevant parameters.

6. Magnetic Sand Castles

Imagine a mixture of standard sand, water, and a small fraction of ferromagnetic micro-particles. When you build a structure (like a sandcastle) and subject it to an external magnetic field, the internal capillary bridges rearrange due to the alignment of the ferromagnetic particles with the field. Investigate how dynamically changing the magnetic field affects the macroscopic stability of the structure. Determine if a critical condition exists for a catastrophic collapse and characterize any hysteresis or memory effects in the system's stability.

7. Coherer Detection



Coherer is an old type of electromagnetic wave detector, which triggers the sharp reduction of the resistance under the influence of the incident radiofrequency wave. Study the conditions at which the coherer can detect the signal, and how it depends on the incident wave amplitude and frequency, the granules size, material and packaging density.

8. Frustrated Klustering

The game Kluster involves placing magnets one at a time on a surface without letting any two of them snap together. Within a fixed area, and with each magnet fixed in place once set down, investigate the stable configurations of N upward- and M downward-polarized magnets. Determine the maximum density for which a stable configuration exists, and how this compares to the maximum density reachable by placing magnets one at a time while keeping every intermediate configuration stable. Investigate how these densities, and the resulting configurations' resilience to perturbations, depend on the ratio N:M, and how allowing the magnets to reorient in-plane changes the achievable density and stability.
https://www.youtube.com/shorts/aAQ3LLttXEs

Physics of Art

9. The Enigma of a Physical Ring

While going through the tapes of the Finnish national archive, an artist found footage of a mysterious physical experiment (https://mikataanila.com/fysikaalinen-rengas-a-physical-ring/). The only remaining information was the footage itself and the name of the tape: "A Physical Ring". The full video documents the setup of the experiment in total silence, but with contributed music, the artist transformed it into a piece of art. 1. Examine the footage and uncover the physical phenomenon (or phenomena) that the authors of the experiment were trying to demonstrate. 2. Recreate the setup presented in the available video and replicate the experiment. Explain relevant phenomena and parametric dependencies.
The attached video is a short demo; final video will be available if problem makes it to the final list: https://drive.google.com/open?id=1pGL8xQsp17l-tf-ODQlyKOPUcjhypKlZ

Physics of materials

10. Spiraling paper
Suppose you have a stack of an arbitrary number of paper sheets. If you push a pen or a stick with a certain force on the top sheet and start to draw circles with a certain angular velocity, the sheets will start to spiral. How does the shape of the spiral change in time, and how does it depend on the number of sheets, the force applied, and the properties of the paper?

11. Net knot propagation
A net with a single torn strand can either have the tear propagate through a large portion of the net, or have it arrested after only a few neighboring strands break, depending on the net's structure. Investigate how the mesh topology and the knot type determine how far a tear propagates from a single torn strand, focusing in particular on whether sliding knots arrest tears better than rigid ones. Investigate whether introducing aperiodicity into the mesh changes this, and whether these effects differ from what is known for rigidly-jointed lattices.

12. Macroscopic quasiparticles
A simple demonstration of lattice vibrations can be a 1D array of coupled spring-mass oscillators. Measure the dispersion of your acoustic and optical phonons. How does the system's behavior change if the masses are replaced by magnets? Does the phonon energy change depend on the magnets' strength and arrangement through the chain? Can you determine and measure an analogue "magnetoelastic coupling" constant? Hint: the less friction, the better!
https://www.youtube.com/watch?v=M4WQs_U1nmU

13. Squeaky Shoes
When you walk in the rain and then go inside, on some surfaces there is an effect, that the shoes make a squeaking sound. What are the relevant parameters for squeaking and how can one maximize/minimize the squeaking sound?

Granular physics

14. Pebbles in boot
When descending a slope covered with loose granular material such as volcanic ash or scree, hikers often find gravel inside their boots even when properly laced. Reproduce this phenomenon under controlled conditions, using a standardized granular material and a fixed boot design driven down a slope at a controlled pace. Investigate how particles enter the boot during downhill walking, and develop and validate a quantitative model for the rate and size distribution of particles that end up inside.

Acoustics

15. Ice Ice Cubey
When we pour water down on ice cubes one can sometimes hear a peculiar cracking sound. Such cracks are rare and quiet when the water is very cold or very hot. Find the dependance of the sound level and frequency of cracking on the ice and water temperatures. What shape of ice cubes produce the loudest cracks and why?

16. Acoustic Camera
A bat locates its prey using only two microphones (its ears). Suppose a human face is illuminated by a single, fixed point source of sound placed 1 meter away, and an array of microphones records the reflected sound. The face and the sound source are stationary. What is the smallest number of microphones needed to acquire enough acoustic information to distinguish one human face from another?

17. Shaken soda syndrome
If two identical cans of a carbonated beverage are rolled down an inclined plane after shaking one of the two, the unshaken one arrives at the bottom faster than the shaken one. Explain the phenomenon and investigate its dependence on the relevant parameters. Can you use this method to determine the precise carbonation content of your soda?

18. Extended Kapitza pendulum
A pendulum whose pivot oscillates vertically can be made to balance stably in its inverted position. This phenomenon, known as dynamical stabilization, was analyzed by Kapitza. The stability criterion and effective potential for a single pendulum under sinusoidal driving are well established. When periodic driving is replaced or supplemented by random noise, the situation becomes theoretically contested: it has been argued both that noise alone can stabilize the inverted position and that it cannot, with the outcome depending sensitively on the spectral character of the noise. Investigate this experimentally and theoretically. Derive how the effective potential of the driven pendulum is modified when the driving contains a stochastic component and determine what properties of the noise govern whether stabilization occurs. Extend your theoretical framework to a system of two or more inverted pendulums sharing a common driven pivot and coupled mechanically. Derive the conditions for stabilization of the coupled system. Measure how the stability boundary depends on the coupling strength and the initial relative configuration of the pendulums. Determine whether the coupled system exhibits any collective behavior that has no counterpart in the single-pendulum case.

Kapitza Pendulum: https://www.youtube.com/watch?v=GgYABmG_bto

Double: https://www.youtube.com/watch?v=gDnbn0Wz_tg

19. Chaotic car
The Malkus waterwheel is an apparatus consisting of leaking cups arranged on the rim of a freely rotating wheel, fed by a constant stream of water from above. Unlike a conventional waterwheel, its rotation is chaotic: it speeds up, slows down, and reverses direction unpredictably. Describe its dynamics and characterize the relevant parameters. Build a vehicle propelled by a Malkus waterwheel that travels as far and/or as fast as possible. https://www.youtube.com/watch?v=Lh1iGSCD8II

20. Sphere Funneling
The trajectory and fall time of a single rolling sphere given a tangential launch speed in a conical funnel is known analytically. Extend this to study how dropping N identical spheres into the funnel at once, each with the same initial tangential speed, changes the inward spiral motion. For a fixed funnel slope, is there a critical number of spheres below which every sphere's exit time and trajectory remain predictable from the initial conditions, and above which system's fine details are effectively unpredictable? How does this number depend on the funnel's slope angle and on the common initial tangential speed. Does a steeper slope or a higher initial speed raise or lower the threshold at which the system becomes unpredictable?

21. Twirling Tea bag
A tea bag is placed in a cup so its string passes over the rim, with the tag hanging freely outside. When the cup is carried by a walking person, the tag may exhibit a variety of motions, including swinging, rotation about the string, and combinations of the former. Investigate and explain the behavior of the tag and determine how its motion depends on walking style and other relevant parameters. Under what conditions is the resulting motion periodic, quasi-periodic, or chaotic? Can you replace walking style with controlled driving to make the system more predictable?

22. Fan Maker
There exist different types of (non-electric) hand fans. Investigate the parameters for an optimal cooling effect (e.g., size, shape, waving frequency, distance). Find the most optimal design - when self-built, it must be one-handed usable.

23. Best Teapot
The quality of a teapot is determined by the fact how transparent and long jets can be poured from the spout of a teapot. As it could be easily found out, the more laminar is the jet, the better is the teapot. Devise the best teapot with the length of the spout limited to 1 cm and the volume of the teapot limited to 300 ml. How will it work for the water of different temperature? https://youtu.be/4R6t4qozgVU

Thermodynamics

24. Jumping Bean
A Mexican jumping bean moves from warmer to cooler regions through a temperature-biased random walk. Construct a passive, non-biological analogue reproducing this behavior, and extend it to a dense collection of many such "beans" placed together in the same temperature gradient. Investigate how interactions between beans affect the ensemble's drift and diffusion compared to a single bean, and determine how the ensemble's migration speed depends on the individual beans' response to temperature and on their density.
https://www.youtube.com/watch?v=La7lajbpj3M

25. Bristle-bot swarm
A "Bristle-bot" is a simple toy made by attaching a vibrating motor to a small brush, causing it to move erratically across the surface. When placing a swarm of Bristle-bots into a confined space, they can start to exhibit collective behavior. Can this collective motion be used to solve a maze more quickly than with the stochastic motion of a single bot? Investigate the efficiency of your maze-solving swarm as a function of bot number, shape, and settings as well as maze design.
https://www.youtube.com/watch?v=jkoXs5HrUjU - A single bristle bot in a maze.

Food physics

26. Causio-e-pepe

When pecorino cheese is stirred into hot water, it can either form a smooth, glossy sauce or collapse into a stringy, separated mass: the "cacio e pepe catastrophe." Recent work has shown that adding starch shifts the boundary between these two outcomes in temperature–composition space. Build a predictive model for when and how fast this transition occurs, rather than only whether it occurs, and test it experimentally. In particular: does the pathway by which the sauce is heated and stirred matter as much as its final temperature and composition? Can your model predict the stabilizing starch fraction for a cheese, or an entire dish, that has not yet been tested? (Other sauces showing similar mechanisms are allowed so that this is not pecorino specific)

Physics of information

27. Bottled bits
A container with two small holes, one on top, one on the bottom, and filled with liquid stops leaking from the bottom hole when the top hole is covered, a well-known effect. Using two holes on top instead of one, the system behaves as a NAND logic gate. Determine the reaction time of this switching process and how it depends on the properties of the liquid and the container, and investigate how to minimize it. Can you extend this idea to build more complex logic gates or circuits, and determine their minimum reaction times?

Optics

28. Singing light
Construct a speaker by exploiting the photo-acoustic effect and characterize its response (sensitivity, sound pressure level and fidelity). What design choices determine the properties and quality of the emitted sound? Is it possible to suppress specific frequencies (i.e. low/high pass filter) by choosing appropriate excitation media? https://www.youtube.com/watch?v=K4LKiJKNsZc

Soap bubbles physics

29. Soap film collisons
Two water drops placed on a horizontal soap film distort the film and attract each other, exhibiting orbiting and merging motion. Explain and predict how the system's fate — the drops scattering apart, being captured into a slowly decaying orbit, or merging immediately — depends on their initial velocity and impact parameter, and determine the merger timescale in the capture regime. Extend this to three or more drops: explain and predict which initial configurations lead to a stable multi-drop arrangement, sequential mergers, or the ejection of one or more drops. https://www.youtube.com/shorts/RZVby7vtrVQ

30. Fluttering soap film
When a uniform stream of air is directed at a soap film held in a rigid frame, the film first bulges into a stable, curved shape, then vibrates once the airflow velocity exceeds a critical threshold. Determine the film's equilibrium shape as a function of the airflow velocity, and investigate the conditions under which this shape becomes unstable. Determine how the critical velocity depends on the relevant parameters, and to what extent this instability follows the same scaling as the flutter of elastic membranes and flags in a flow.

Fluid dynamics & hydrodynamics

31. Aerodynamic Juggler
A light ball may be suspended on a fast vertical air stream. With enough care, one can place a second ball on the same air stream, and both remain floating, dancing around each other. Investigate the conditions for the phenomenon to be stable in the multi-ball regime. Can you create a setup in which you could add more balls to the air stream without destabilizing the system? How many balls can you stack? How does tilting the air stream or changing the shape of the ball affect your solution?

32. Water wire
Suppose you have a wire connecting two points in space, and you have a water stream flowing from the higher point along the wire (example: here). What is the optimal shape of the wire for the water to reach the biggest horizontal distance without dripping down completely, for a given height difference? How does the optimal shape depend on the flow rate of the water, and on the properties of the wire (e.g. its roughness and radius)? You may choose a malleable material for your wire to produce arbitrary shapes.

33. Ink Boat
A drop of common ballpoint-pen ink applied to a porous carrier, such as a leaf, propels it across a water surface. Unlike a bare droplet, the carrier releases ink gradually, so the surface-tension gradient driving the motion evolves over the course of the run. Investigate how the carrier's material properties (porosity, shape, wetted area) govern this release rate, and how the ink patterns left on the water surface in the carrier's wake (their spacing, and how the pattern changes or fades along the trail) encode that underlying release dynamics. Determine to what extent the trail pattern alone is related to the maximal velocity achievable for such a system.
https://www.youtube.com/watch?v=gNoWqvIC4Ks

34. Thermohaline fingers
When a warm, saline layer rests above a cool, fresh layer, the system appears gravitationally stable yet spontaneously develops elongated convective structures that transport heat and solute across the interface at markedly different rates. Investigate this instability. Measure how finger geometry and growth rate depend on the temperature difference and salinity contrast across the interface. Determine the flux ratio - the ratio of the buoyancy flux carried by solute to that carried by heat - and establish how it varies with the stratification. Under what conditions does the finger-dominated regime give way to bulk overturning? The ratio of solute diffusivity to thermal diffusivity differs by order of magnitude between ocean water and a stellar interior. Using dimensional analysis and your experimental results, extrapolate how finger geometry and mix efficiency scale into these extreme regimes, and discuss the physical consequences. https://www.youtube.com/watch?v=_uGqCLIOnPU

35. It's raining
A "cloud" can be created in a jar by pouring in hot water and cooling the air with an ice-topped lid. The basic condensation mechanism is well understood. Under a fixed, reproducible setup, investigate the conditions under which the condensed droplets grow large enough to fall as visible "rain" rather than staying suspended as fog, and determine how the resulting droplet size and fall behavior depend on these parameters. Separately, investigate whether the droplets can instead be made to fall as ice crystals ("snow") and determine what's needed to reliably produce one outcome or the other. https://www.youtube.com/shorts/0EmEcWKUoJ0

36. Viscous rolling
A jar partly filled with a viscous liquid, together with a ping-pong ball, exhibits periodic ("rock-and-roll") motion when rolled down an inclined slope. Periodic descent of liquid-containing rolling objects is well documented for liquid-only bottles and for a solid cylinder rolling inside a viscous-fluid-filled shell. How does the addition of a ball change this periodic motion? Develop a model relating the motion (period, jerk amplitude, average descent speed) to the ball's density relative to the liquid and to the other relevant parameters (incline angle, liquid viscosity, fill level, ball size). Can the system's parameters be tuned to

produce a specific target motion profile, and what range of motion profiles is achievable this way? [Video from (1:58 - 2:43): https://youtu.be/72DCj3BztG4?is=Vres9QmdtDMtBGyS]

37. Not-So-Precious energy

When a shaken carbonated drink like soda is opened, the sudden pressure drop causes previously nucleated gas bubbles to rapidly expand, ejecting liquid. It is claimed that rotating the can beforehand can prevent this. Investigate this claim, distinguishing between different possible rotation motions, and compare their effectiveness to simply letting the shaken can rest undisturbed for a period of time. Are there other ways to prevent the liquid from being ejected upon opening?
https://www.youtube.com/watch?v=AHjbafEl9xw

38. Sandstorms



Wind blowing over dry sand creates ripple patterns. Investigate how the pattern's wavelength, shape, and coarsening depend on a wind that varies in speed and/or direction rather than blowing steadily, and explain the two-dimensional structure of the pattern. Investigate and explain what changes in the resulting pattern when the sand is slowly made wet.

39. Fountain in a pot

On the video you may observe a drop periodically flying high over the water level in the pot. How does the drop's size, altitude and its period of formation depend on relevant parameters? Which jet, pot and fluid parameters can be determined by observing just these drops? https://youtu.be/2q1EPAacsUA